

The background of the slide features a large, faint circular logo of Tianjin University in the upper right, containing the university's name in English ('TIANJIN UNIVERSITY') and Chinese ('天津大学'), along with the founding year '1895'. In the lower left, there is a faint line drawing of a traditional Chinese building with a tiled roof. The main title is centered on the right side of the slide.

操作系统结构

Operating System Structures

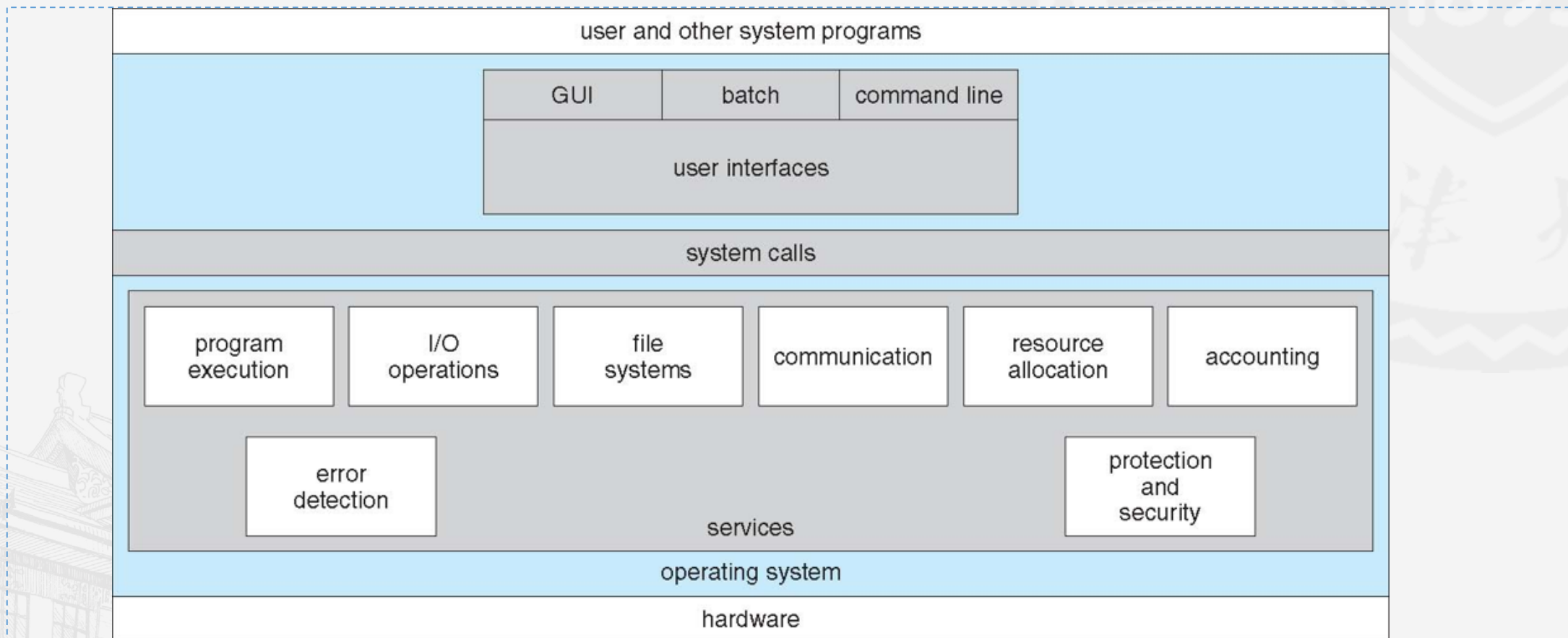


主要内容

- 操作系统所提供的功能
- 系统调用 System call
- 库函数 C library
- 典型的UNIX操作系统架构



操作系统提供的服务





操作系统结构

Operating System Structures

系统调用示例

EXAMPLE OF STANDARD API

As an example of a standard API, consider the `read()` function that is available in UNIX and Linux systems. The API for this function is obtained from the `man` page by invoking the command

```
man read
```

on the command line. A description of this API appears below:

```
#include <unistd.h>

ssize_t read(int fd, void *buf, size_t count)
```

ssize_t	read	(int fd, void *buf, size_t count)
return value	function name	parameters

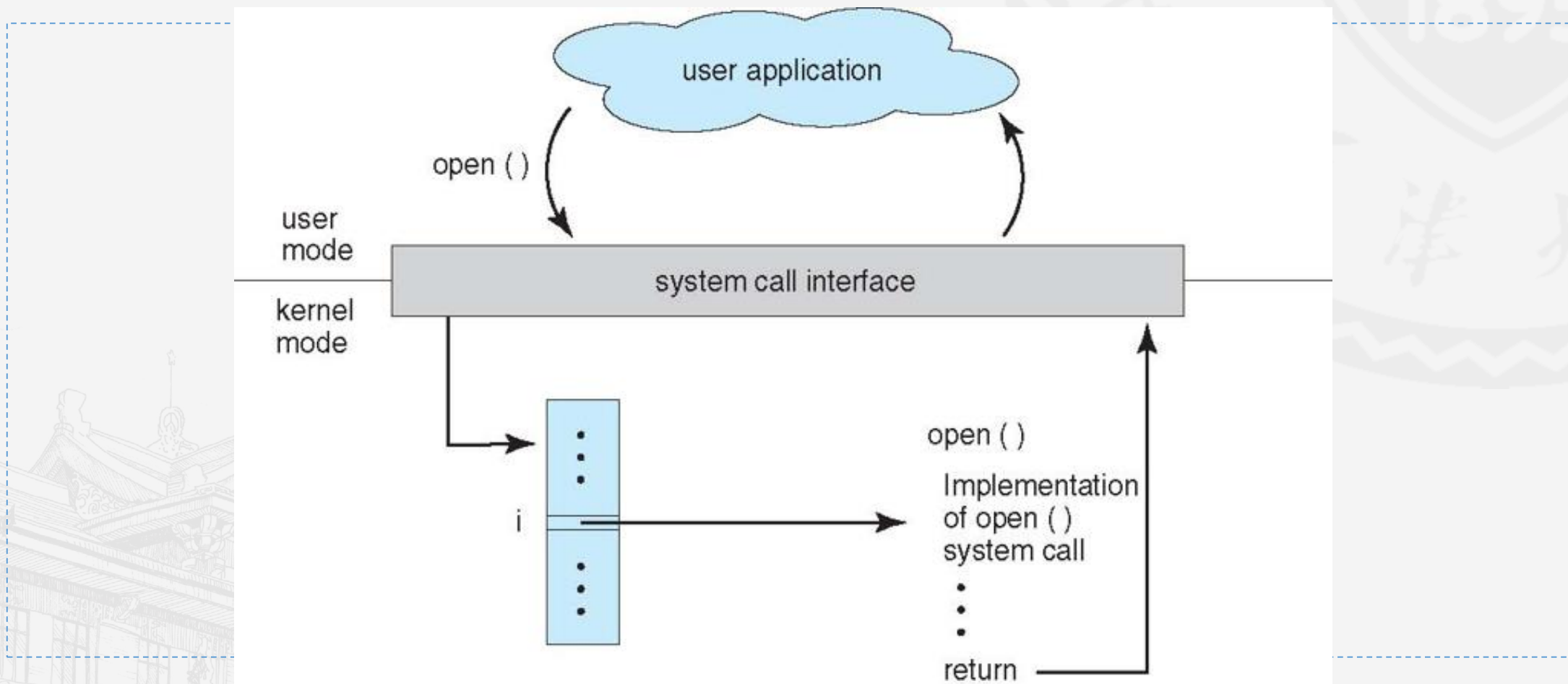
A program that uses the `read()` function must include the `unistd.h` header file, as this file defines the `ssize_t` and `size_t` data types (among other things). The parameters passed to `read()` are as follows:

- `int fd`—the file descriptor to be read
- `void *buf`—a buffer where the data will be read into
- `size_t count`—the maximum number of bytes to be read into the buffer

On a successful read, the number of bytes read is returned. A return value of 0 indicates end of file. If an error occurs, `read()` returns `-1`.

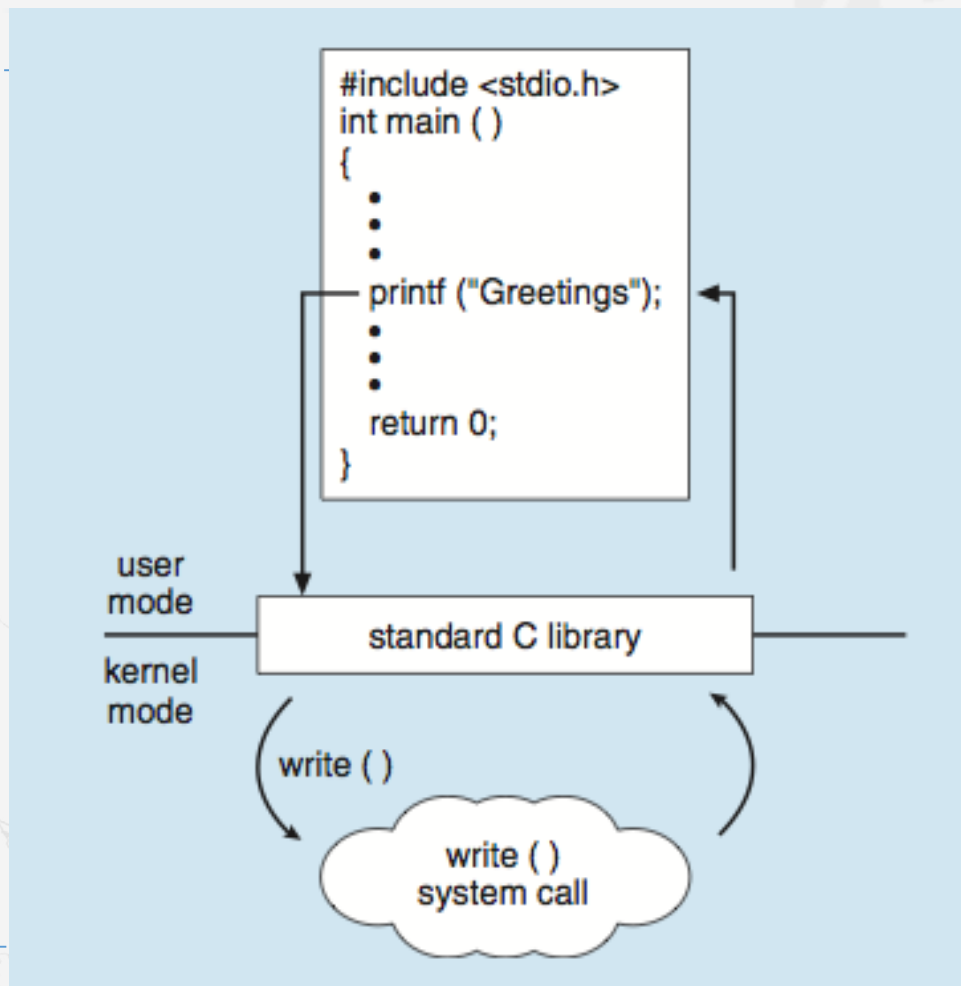


系统调用 (API - System Call)



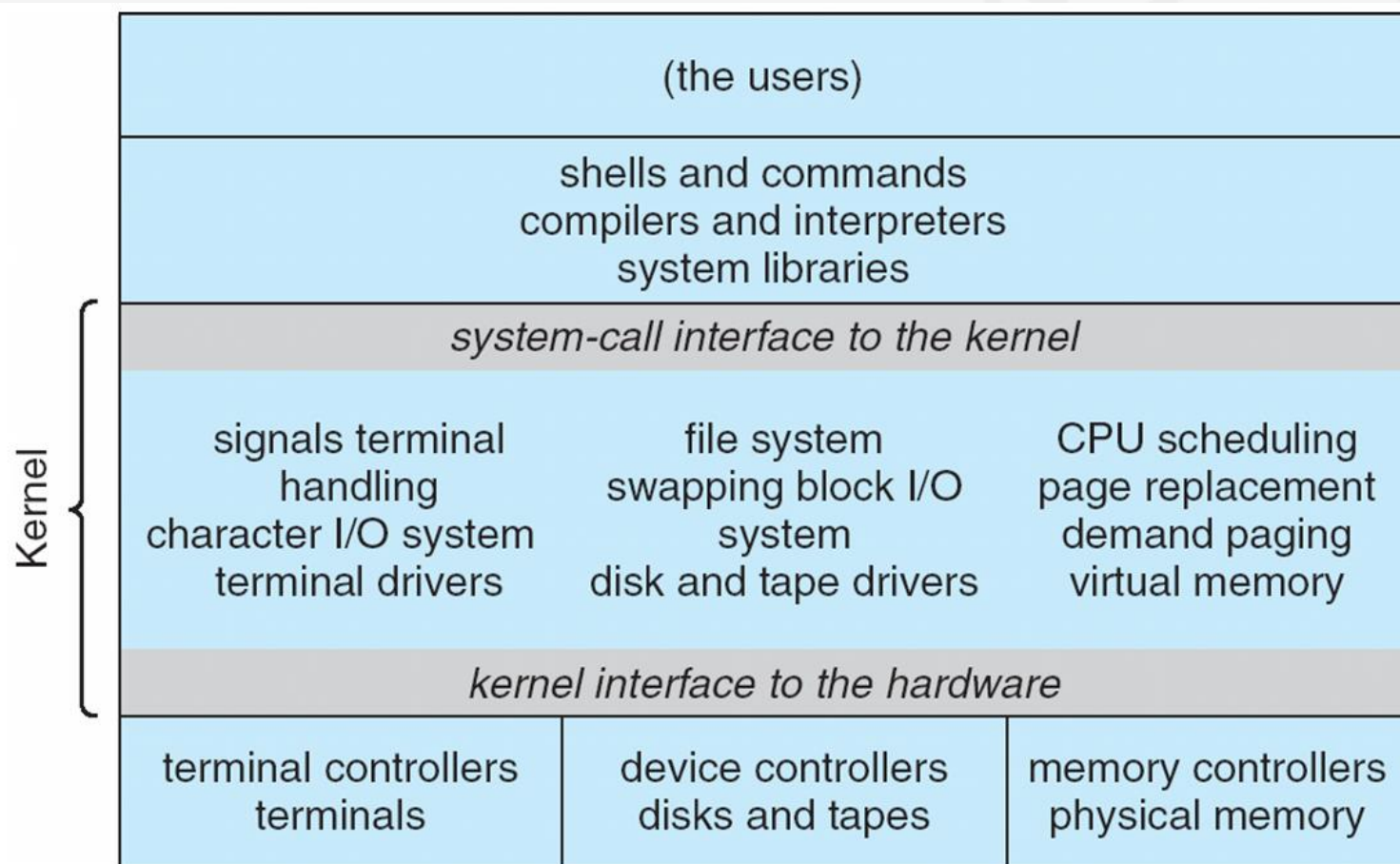


标准 C 库函数





UNIX 系统架构





课后阅读

- 《操作系统概念 OSC（第9版）》第2章。
- 《现代操作系统 MOS（第4版）》第1章。